

Online Feedback Collector - Project Documentation

Project Overview

A full-stack web application built with Flask that enables users to submit feedback through an intuitive form and provides administrators with a comprehensive dashboard for data analysis and management.

Core Purpose

User Experience: Simple, responsive feedback submission interface

Admin Analytics: Real-time statistics, charts, and data export capabilities

Data Management: Secure storage and retrieval of feedback data

Architecture

Backend (Flask)

Framework: Flask 2.3.3 with Python 3

Database: SQLite for lightweight, serverless data storage

Authentication: Session-based admin access control

API: RESTful endpoints for data operations

Frontend

UI Framework: Bootstrap 5 for responsive design

Templates: Jinja2 templating engine

Interactivity: JavaScript with Chart.js for data visualization

Styling: Custom CSS with modern animations

Database Schema

Feedback Table

```
CREATE TABLE feedback (
```

```
id INTEGER PRIMARY KEY AUTOINCREMENT,  
name TEXT NOT NULL,  
email TEXT NOT NULL,  
rating INTEGER NOT NULL CHECK(rating >= 1 AND rating <= 5),  
comments TEXT,  
date_submitted TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Users Table

```
CREATE TABLE users (  
id INTEGER PRIMARY KEY AUTOINCREMENT,  
name TEXT NOT NULL,  
email TEXT UNIQUE NOT NULL,  
password_hash TEXT NOT NULL,  
is_admin BOOLEAN DEFAULT 0,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Key Features

User Interface

Feedback Form: Name, email, 1-5 star rating, optional comments

Real-time Validation: Client-side form validation with visual feedback

Responsive Design: Mobile-first approach with Bootstrap 5

Success Messaging: Clear feedback confirmation

Admin Dashboard

Authentication: Secure login for authorized administrators

Statistics Panel: Total feedback, average rating, response rates

Interactive Charts:

Rating Distribution: Pie chart showing feedback breakdown

Daily Feedback Trends: Bar chart displaying last 7 days of activity

Data Management: Sortable table with all feedback entries

Export Functionality: CSV download for external analysis

Real-time Updates: Dynamic data refresh capabilities

Technical Implementation

Core Routes

GET / - Home page with feedback form

POST /submit-feedback - Feedback submission endpoint

GET/POST /login - Admin authentication

GET /admin-dashboard - Protected admin interface

GET /export-csv - Data export functionality

GET /chart-data - JSON API for chart data

GET /logout - Session termination

Security Features

Input Validation: Both client and server-side validation

SQL Injection Prevention: Parameterized queries

XSS Protection: Jinja2 auto-escaping

Session Management: Secure Flask sessions

Admin Access Control: Decorator-based route protection

Data Flow

1. User Submission: Form → Validation → Database storage → Success response
2. Admin Access: Login → Session creation → Dashboard access → Data visualization
3. Export: Database query → CSV generation → File download

Project Structure

OnlineFeedbackCollector/

- ├── app.py # Main Flask application
- ├── requirements.txt # Python dependencies
- ├── database.db # SQLite database
- ├── static/
 - | ├── css/style.css # Custom styling
 - | └── js/script.js # Frontend JavaScript
- ├── templates/
 - | ├── layout.html # Base template
 - | ├── index.html # Home page
 - | ├── login.html # Admin login
 - | └── admin.html # Admin dashboard
- └── README.md # Detailed setup guide

Dependencies

Flask 2.3.3: Web framework

Werkzeug 2.3.7: WSGI utilities

Jinja2 3.1.2: Template engine

SQLite: Database (built-in Python)

Design Principles

User-Centric: Intuitive interface with clear visual hierarchy

Responsive: Mobile-first responsive design

Accessible: Semantic HTML5 with proper ARIA labels

Performance: Optimized queries and efficient data handling

Security: Multi-layer validation and protection mechanisms

Analytics & Reporting

Real-time Statistics: Live feedback metrics

Trend Analysis: 7-day activity tracking

Rating Distribution: Visual breakdown of user satisfaction

Data Export: CSV format for external analysis tools

Authentication System

Admin Users: Predefined email-based access (tharun120@admin.com, sudheer@admin.com)

Session Management: Secure session handling with logout functionality

Route Protection: Decorator-based access control for admin areas

Deployment Considerations

Development: Built-in Flask development server

Production Ready: Compatible with WSGI servers (Gunicorn, uWSGI)

Scalability: SQLite suitable for small to medium applications

Portability: Self-contained with minimal external dependencies

Business Value

Customer Insights: Direct feedback collection and analysis

Decision Support: Data-driven insights from user ratings

Operational Efficiency: Automated feedback management

User Engagement: Simple, accessible feedback mechanism

Developed with Flask, Bootstrap, and modern web technologies for optimal user experience and administrative control.